

Deliang Fan

associate professor

基本信息

姓名: Deliang Fan

所属院校: 亚利桑那州立大学

地区: United States

年龄范围: 30岁至40岁

海外经历: 有

学科领域

人工智能

电子与计算机工程

集成电路设计与集成系统

信息安全

研究领域

深度学习与神经网络

计算机体系结构

存储器设计与架构

低功耗设计与能耗优化

个人简介

Deliang Fan is an Associate Professor in Electrical Engineering at the School of Electrical, Computer and Energy Engineering, Arizona State University. He holds a PhD in electrical and computer engineering from Purdue University (2015). His expertise lies in efficient AI hardware and algorithms, digital chip design, and secure and trustworthy AI systems. After working at Johns Hopkins University, he's back at ASU. He teaches EEE 334 Circuits II and plans to teach a course on AI computing hardware. A prolific researcher, he has over 170 papers, 15 patents, and has reviewed for 30+ journals/conferences. He has won multiple awards, including a 2022 NSF CAREER Award. In his free time, he enjoys tennis and swimming.

工作经历

Arizona State University - Associate Professor

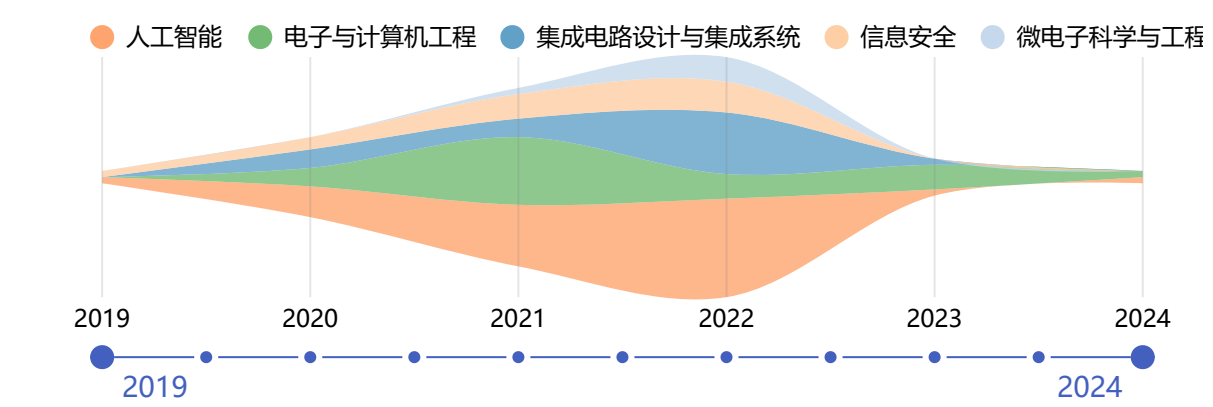
2022-08 -

Johns Hopkins University - Associate Professor	2023-08 - 2024-08
Arizona State University - Assistant Professor	2019-08 - 2022-08
University of Central Florida - assistant professor	2015-08 - 2019-08

教育经历

Purdue University - PHD	2013-01 - 2015-08
Purdue University - MS	2010-08 - 2012-12

近年发表的论文



A Progressive Subnetwork Searching Framework for Dynamic Inference

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Yang, Li; He, Zhezhi; Cao, Yu; Fan, Deliang

被引用次数： 12024

Energy Efficient Computing With High-Density, Field-Free STT-Assisted SOT-MRAM (SAS-MRAM)

IEEE TRANSACTIONS ON MAGNETICS

Hwang, William; Xue, Fen; Zhang, Fan; Song, Ming-Yuan; Lee, Chien-Min; Turgut, Emrah; Chen, T. C.; Bao, Xinyu; Tsai, Wilman; Fan, Deliang; Wang, Shan X.

被引用次数： 62023

Aligner-D: Leveraging In-DRAM Computing to Accelerate DNA Short Read Alignment

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Zhang, Fan; Angizi, Shaahin; Sun, Jiao; Zhang, Wei; Fan, Deliang

被引用次数： 42023

FP-IMC: A 28nm All-Digital Configurable Floating-Point In-Memory Computing Macro

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Saikia, Jyotishman; Sridharan, Amitesh; Yeo, Injune; Venkataramanaiah, Shreyas; Fan, Deliang; Seo, Jae-Sun

被引用次数: 3

2023

A 65nm RRAM Compute-in-Memory Macro for Genome Sequencing Alignment

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Zhang, Fan; He, Wangxin; Yeo, Injune; Liehr, Maximilian; Cady, Nathaniel; Cao, Yu; Seo, Jae-sun; Fan, Deliang

被引用次数: 1

2023

DSPIMM: A Fully Digital SParse In-Memory Matrix Vector Multiplier for Communication Applications

Design Automation Conference

Amitesh Sridharan; Fan Zhang; Yang Sui; Bo Yuan; Deliang Fan

被引用次数: 0

2023

Sparse and Robust RRAM-based Efficient In-memory Computing for DNN Inference

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Meng, Jian; Yeo, Injune; Shim, Wonbo; Yang, Li; Fan, Deliang; Yu, Shimeng; Seo, Jae-Sun

被引用次数: 2

2022

Improving the accuracy and robustness of RRAM-based in-memory computing against RRAM hardware noise and adversarial attacks

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Cherupally, Sai Kiran; Meng, Jian; Rakin, Adnan Siraj; Yin, Shihui; Yeo, Injune; Yu, Shimeng; Fan, Deliang; Seo, Jae-Sun

被引用次数: 5

2022

Improving DNN Hardware Accuracy by In-Memory Computing Noise Injection

IEEE DESIGN & TEST

Cherupally, Sai Kiran; Meng, Jian; Rakin, Adnan Siraj; Yin, Shihui; Seok, Mingoo; Fan, Deliang; Seo, Jae-Sun

被引用次数: 8

2022

ResSFL: A Resistance Transfer Framework for Defending Model Inversion Attack in Split Federated Learning

2022 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)

Li, Jingtao; Rakin, Adnan Siraj; Chen, Xing; He, Zhezhi; Fan, Deliang; Chakrabarti, Chaitali

被引用次数: 36

2022

DeepSteal: Advanced Model Extractions Leveraging Efficient Weight Stealing in Memories

43RD IEEE SYMPOSIUM ON SECURITY AND PRIVACY (SP 2022)

Rakin, Adnan Siraj; Chowdhury, Md Hafizul Islam; Yao, Fan; Fan, Deliang

被引用次数: 45

2022

DA³: Dynamic Additive Attention Adaption for Memory-Efficient On-Device Multi-Domain Learning

IEEE/CVF Computer Vision and Pattern Recognition Conference

Li Yang; Adnan Siraj Rakin; Deliang Fan

被引用次数: 0

2022

RepNet: Efficient On-Device Learning via Feature Reprogramming

IEEE/CVF Computer Vision and Pattern Recognition Conference

Li Yang; Adnan Siraj Rakin; Deliang Fan

被引用次数: 0

2022

ResSFL: A Resistance Transfer Framework for Defending Model Inversion Attack in Split Federated Learning

IEEE/CVF Computer Vision and Pattern Recognition Conference

Jingtao Li; Adnan Siraj Rakin; Xing Chen; Zhezhi He; Deliang Fan; Chaitali Chakrabarti

被引用次数: 0

2022

Contrastive Dual Gating: Learning Sparse Features With Contrastive Learning

IEEE/CVF Computer Vision and Pattern Recognition Conference

Jian Meng; Li Yang; Jinwoo Shin; Deliang Fan; Jae-Sun Seo

被引用次数: 0

2022

Gradient-Based Novelty Detection Boosted by Self-Supervised Binary Classification

THIRTY-SIXTH AAAI CONFERENCE ON ARTIFICIAL INTELLIGENCE / THIRTY-FOURTH CONFERENCE ON INNOVATIVE APPLICATIONS OF ARTIFICIAL INTELLIGENCE / TWELVETH SYMPOSIUM ON EDUCATIONAL ADVANCES IN ARTIFICIAL INTELLIGENCE

Sun, Jingbo; Yang, Li; Zhang, Jiaxin; Liu, Frank; Halappanavar, Mahantesh; Fan, Deliang; Cao, Yu

被引用次数: 5

2022

Efficient Multi-task Adaption for Crossbar-based In-Memory Computing

2022 56TH ASIOMAR CONFERENCE ON SIGNALS, SYSTEMS, AND COMPUTERS

Zhang, Fan; Yang, Li; Fan, Deliang

被引用次数: 0

2022

DeepSteal: Advanced Model Extractions Leveraging Efficient Weight Stealing in Memories

IEEE Symposium on Security and Privacy

Adnan Siraj Rakin; Md Hafizul Islam Chowdhury; Fan Yao; Deliang Fan

被引用次数: 0

2022

Gradient-Based Novelty Detection Boosted by Self-Supervised Binary Classification

AAAI Conference on Artificial Intelligence

Jingbo Sun; Li Yang; Jiaxin Zhang; Frank Liu; Mahantesh Halappanavar; Deliang Fan; Yu Cao

被引用次数: 0

2022

Sparse and Robust RRAM-based Efficient In-memory Computing for DNN Inference

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Meng, Jian; Yeo, Injune; Yang, Li; Fan, Deliang; Seo, Jae-sun; Yu, Shimeng; Shim, Wonbo

被引用次数: 0

2022

XMA2: A crossbar-aware multi-task adaption framework via 2-tier masks

FRONTIERS IN ELECTRONICS

Zhang, Fan; Yang, Li; Meng, Jian; Seo, Jae-sun; Cao, Yu; Fan, Deliang	2022
被引用次数: 2	
XBM: A Crossbar Column-wise Binary Mask Learning Method for Efficient Multiple Task Adaption	
27TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2022	
Zhang, Fan; Yang, Li; Meng, Jian; Cao, Yu (Kevin); Seo, Jae-sun; Fan, Deliang	
被引用次数: 2	2022
Beyond Not-Forgetting: Continual Learning with Backward Knowledge Transfer	
ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS 35 (NEURIPS 2022)	
Lin, Sen; Yang, Li; Fan, Deliang; Zhang, Junshan	
被引用次数: 10	2022
Get More at Once: Alternating Sparse Training with Gradient Correction	
ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS 35 (NEURIPS 2022)	
Yang, Li; Meng, Jian; Seo, Jae-Sun; Fan, Deliang	
被引用次数: 0	2022
XMA: A Crossbar-aware Multi-task Adaption Framework via Shift-based Mask Learning Method	
PROCEEDINGS OF THE 59TH ACM/IEEE DESIGN AUTOMATION CONFERENCE, DAC 2022	
Zhang, Fan; Yang, Li; Meng, Jian; Seo, Jae -Sun; Cao, Yu (Kevin); Fan, Deliang	
被引用次数: 0	2022
Rep-Net: Efficient On-Device Learning via Feature Reprogramming	
2022 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)	
Yang, Li; Rakin, Adnan Siraj; Fan, Deliang	
被引用次数: 9	2022
XMA: a crossbar-aware multi-task adaption framework via shift-based mask learning method	
Design Automation Conference	
Fan Zhang; Li Yang; Jian Meng; Jae-sun Seo; Yu (Kevin) Cao; Deliang Fan	
被引用次数: 0	2022
MeF-RAM: A New Non-Volatile Cache Memory Based on Magneto-Electric FET	
ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS	
Angizi, Shaahin; Khoshavi, Navid; Marshall, Andrew; Dowben, Peter; Fan, Deliang	
被引用次数: 7	2022
Contrastive Dual Gating: Learning Sparse Features With Contrastive Learning	
2022 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)	
Meng, Jian; Yang, Li; Shin, Jinwoo; Fan, Deliang; Seo, Jae-Sun	
被引用次数: 3	2022
Non-Structured DNN Weight Pruning--Is It Beneficial in Any Platform?	

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Ma, Xiaolong; Lin, Sheng; Ye, Shaokai; He, Zhezhi; Zhang, Linfeng; Yuan, Geng; Tan, Sia Huat; Li, Zhengang; Fan, Deliang; Qian, Xuehai; Lin, Xue; Ma, Kaisheng; Wang, Yanzhi

被引用次数: 43

2022

Temperature-Resilient RRAM-Based In-Memory Computing for DNN Inference

IEEE MICRO

Meng, Jian; Shim, Wonbo; Yang, Li; Yeo, Injune; Fan, Deliang; Yu, Shimeng; Seo, Jae-sun

被引用次数: 13

2022

T-BFA: Targeted Bit-Flip Adversarial Weight Attack

IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE

Rakin, Adnan Siraj; He, Zhezhi; Li, Jingtao; Yao, Fan; Chakrabarti, Chaitali; Fan, Deliang

被引用次数: 16

2022

Hybrid RRAM/SRAM in-Memory Computing for Robust DNN Acceleration

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Krishnan, Gokul; Wang, Zhenyu; Yeo, Injune; Yang, Li; Meng, Jian; Liehr, Maximilian; Joshi, Rajiv, V; Cady, Nathaniel C.; Fan, Deliang; Seo, Jae-Sun; Cao, Yu

被引用次数: 9

2022

Exploring Model Stability of Deep Neural Networks for Reliable RRAM-Based In-Memory Acceleration

IEEE TRANSACTIONS ON COMPUTERS

Krishnan, Gokul; Yang, Li; Sun, Jingbo; Hazra, Jubin; Du, Xiacong; Liehr, Maximilian; Li, Zheng; Beckmann, Karsten; Joshi, Rajiv, V; Cady, Nathaniel C.; Fan, Deliang; Cao, Yu

被引用次数: 4

2022

Deep-Dup: An Adversarial Weight Duplication Attack Framework to Crush Deep Neural Network in Multi-Tenant FPGA

PROCEEDINGS OF THE 30TH USENIX SECURITY SYMPOSIUM

Rakin, Adnan Siraj; Luo, Yukui; Xu, Xiaolin; Fan, Deliang

被引用次数: 30

2021

Characterization and Mitigation of Relaxation Effects on Multi-level RRAM based In-Memory Computing

2021 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

He, Wangxin; Shim, Wonbo; Yin, Shihui; Sun, Xiaoyu; Fan, Deliang; Yu, Shimeng; Seo, Jae-sun

被引用次数: 14

2021

KSM: Fast Multiple Task Adaption via Kernel-wise Soft Mask Learning

2021 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION, CVPR 2021

Yang, Li; He, Zhezhi; Zhang, Junshan; Fan, Deliang

被引用次数: 5

2021

RNSiM: Efficient Deep Neural Network Accelerator Using Residue Number Systems

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Roohi, Arman; Taheri, MohammadReza; Angizi, Shaahin; Fan, Deliang	2021
被引用次数: 8	
MetaGater: Fast Learning of Conditional Channel Gated Networks via Federated Meta-Learning	
2021 IEEE 18TH INTERNATIONAL CONFERENCE ON MOBILE AD HOC AND SMART SYSTEMS (MASS 2021)	
Lin, Sen; Yang, Li; He, Zhezhi; Fan, Deliang; Zhang, Junshan	
被引用次数: 5	2021
Leveraging Noise and Aggressive Quantization of In-Memory Computing for Robust DNN Hardware Against Adversarial Input and Weight Attacks	
2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)	
Cherupally, Sai Kiran; Rakin, Adnan Siraj; Yin, Shihui; Seok, Mingoo; Fan, Deliang; Seo, Jae-sun	
被引用次数: 8	2021
Max-PIM: Fast and Efficient Max/Min Searching in DRAM	
2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)	
Zhang, Fan; Angizi, Shaahin; Fan, Deliang	
被引用次数: 7	2021
PIM-Quantifier: A Processing-in-Memory Platform for mRNA Quantification	
2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)	
Zhang, Fan; Angizi, Shaahin; Fahmi, Naima Ahmed; Zhang, Wei; Fan, Deliang	
被引用次数: 11	2021
NeurObfuscator: A Full-stack Obfuscation Tool to Mitigate Neural Architecture Stealing	
2021 IEEE INTERNATIONAL SYMPOSIUM ON HARDWARE ORIENTED SECURITY AND TRUST (HOST)	
Li, Jingtao; He, Zhezhi; Rakin, Adnan Siraj; Fan, Deliang; Chakrabarti, Chaitali	
被引用次数: 11	2021
PIM-Quantifier: A Processing-in-Memory Platform for mRNA Quantification	
Design Automation Conference	
Fan Zhang; Shaahin Angizi; Naima Ahmed Fahmi; Wei Zhang; Deliang Fan	
被引用次数: 0	2021
Max-PIM: Fast and Efficient Max/Min Searching in DRAM	
Design Automation Conference	
Fan Zhang; Shaahin Angizi; Deliang Fan	
被引用次数: 0	2021
Leveraging Noise and Aggressive Quantization of In-Memory Computing for Robust DNN Hardware Against Adversarial Input and Weight Attacks	
Design Automation Conference	
Sai Kiran Cherupally; Adnan Siraj Rakin; Shihui Yin; Mingoo Seok; Deliang Fan; Jae-sun Seo	
被引用次数: 0	2021
Resilient and Secure Hardware Devices Using ASL	

Non-Volatile Approximate Arithmetic Circuits Using Scalable Hybrid Spin-CMOS Majority Gates

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jiang, Honglan; Angizi, Shaahin; Fan, Deliang; Han, Jie; Liu, Leibo

被引用次数: 14

2021

Reducing SRAM Reading Power With Column Data Segment and Weights Correlation Enhancement for CNN Processing

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Xu, Han; Li, Ziwei; Li, Zirui; Fan, Deliang; Qiao, Fei; Wei, Qi; Luo, Li; Liu, Xinjun; Yang, Huazhong

被引用次数: 1

2021

Structured Pruning of RRAM Crossbars for Efficient In-Memory Computing Acceleration of Deep Neural Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Meng, Jian; Yang, Li; Peng, Xiaochen; Yu, Shimeng; Fan, Deliang; Seo, Jae-Sun

被引用次数: 29

2021

Deep-Dup: An Adversarial Weight Duplication Attack Framework to Crush Deep Neural Network in Multi-Tenant FPGA

USENIX Security Symposium

Adnan Siraj Rakin; Yukui Luo; Xiaolin Xu; Deliang Fan

被引用次数: 0

2021

DeepHammer: Depleting the Intelligence of Deep Neural Networks through Targeted Chain of Bit Flips

USENIX Security Symposium

Fan Yao; Adnan Siraj Rakin; Deliang Fan

被引用次数: 0

2020

Network-based multi-task learning models for biomarker selection and cancer outcome prediction

BIOINFORMATICS

Wang, Zhibo; He, Zhezhi; Shah, Milan; Zhang, Teng; Fan, Deliang; Zhang, Wei

被引用次数: 12

2020

Handling Stuck-at-Fault Defects Using Matrix Transformation for Robust Inference of DNNs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, Baogang; Uysal, Necati; Fan, Deliang; Ewetz, Rickard

被引用次数: 26

2020

Harmonious Coexistence of Structured Weight Pruning and Ternarization for Deep Neural Networks

AAAI Conference on Artificial Intelligence

Li Yang; Zhezhi He; Deliang Fan

被引用次数: 0

2020

Sparse BD-Net: A Multiplication-less DNN with Sparse Binarized Depth-wise Separable Convolution

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

He, Zhezhi; Yang, Li; Angizi, Shaahin; Rakin, Adnan Siraj; Fan, Deliang

被引用次数: 16

2020

Non-uniform DNN Structured Subnets Sampling for Dynamic Inference

Design Automation Conference

Li Yang; Zhezhi He; Yu Cao; Deliang Fan

被引用次数: 0

2020

Defending Bit-Flip Attack through DNN Weight Reconstruction

Design Automation Conference

Jingtao Li; Adnan Siraj Rakin; Yan Xiong; Liangliang Chang; Zhezhi He; Deliang Fan; Chaitali Chakrabarti

被引用次数: 0

2020

PIM-Assembler: A Processing-in-Memory Platform for Genome Assembly

Design Automation Conference

Shaahin Angizi; Naima Ahmed Fahmi; Wei Zhang; Deliang Fan

被引用次数: 0

2020

2D MoS₂-Based Threshold Switching Memristor for Artificial Neuron

IEEE ELECTRON DEVICE LETTERS

Dev, Durjoy; Krishnaprasad, Adithi; Shawkat, Mashiyat S.; He, Zhezhi; Das, Sonali; Fan, Deliang; Chung, Hee-Suk; Jung, Yeonwoong; Roy, Tania

被引用次数: 85

2020

Hybrid Spin-CMOS Polymorphic Logic Gate With Application in In-Memory Computing

IEEE TRANSACTIONS ON MAGNETICS

Angizi, Shaahin; He, Zhezhi; Chen, An; Fan, Deliang

被引用次数: 18

2020

Bit-Flip Attack: Crushing Neural Network With Progressive Bit Search

International Conference on Computer Vision

Adnan Siraj Rakin; Zhezhi He; Deliang Fan

被引用次数: 0

2019

T-BFA: Targeted Bit-Flip Adversarial Weight Attack
TPAMI

Adnan Siraj Rakin; Zhezhi He; Jingtao Li; Fan Yao; Chaitali Chakrabarti; Deliang Fan

被引用次数: 0

Hybrid RRAM/SRAM in-Memory Computing for Robust DNN Acceleration

TCAD

Gokul Krishnan; Zhenyu Wang; Injune Yeo; Li Yang; Jian Meng; Maximilian Liehr; Rajiv V. Joshi; Nathaniel C. Cady; Deliang Fan; Jae-Sun Seo; Yu Cao

被引用次数: 0

Defending and Harnessing the Bit-Flip Based Adversarial Weight Attack

CVPR

Zhezhi He; Adnan Siraj Rakin; Jingtao Li; Chaitali Chakrabarti; Deliang Fan

被引用次数: 0

TBT: Targeted Neural Network Attack With Bit Trojan

CVPR

Adnan Siraj Rakin; Zhezhi He; Deliang Fan

被引用次数: 0

KSM: Fast Multiple Task Adaption via Kernel-wise Soft Mask Learning

CVPR

Li Yang; Zhezhi He; Junshan Zhang; Deliang Fan

被引用次数: 0

Handling Stuck-at-Fault Defects Using Matrix Transformation for Robust Inference of DNNs

TCAD

Baogang Zhang; Necati Uysal; Deliang Fan; Rickard Ewetz

被引用次数: 0

Reducing SRAM Reading Power With Column Data Segment and Weights Correlation Enhancement for CNN Processing

TCAD

Han Xu; Ziwei Li; Ziru Li; Deliang Fan; Fei Qiao; Qi Wei; Li Luo; Xinjun Liu; Huazhong Yang

被引用次数: 0

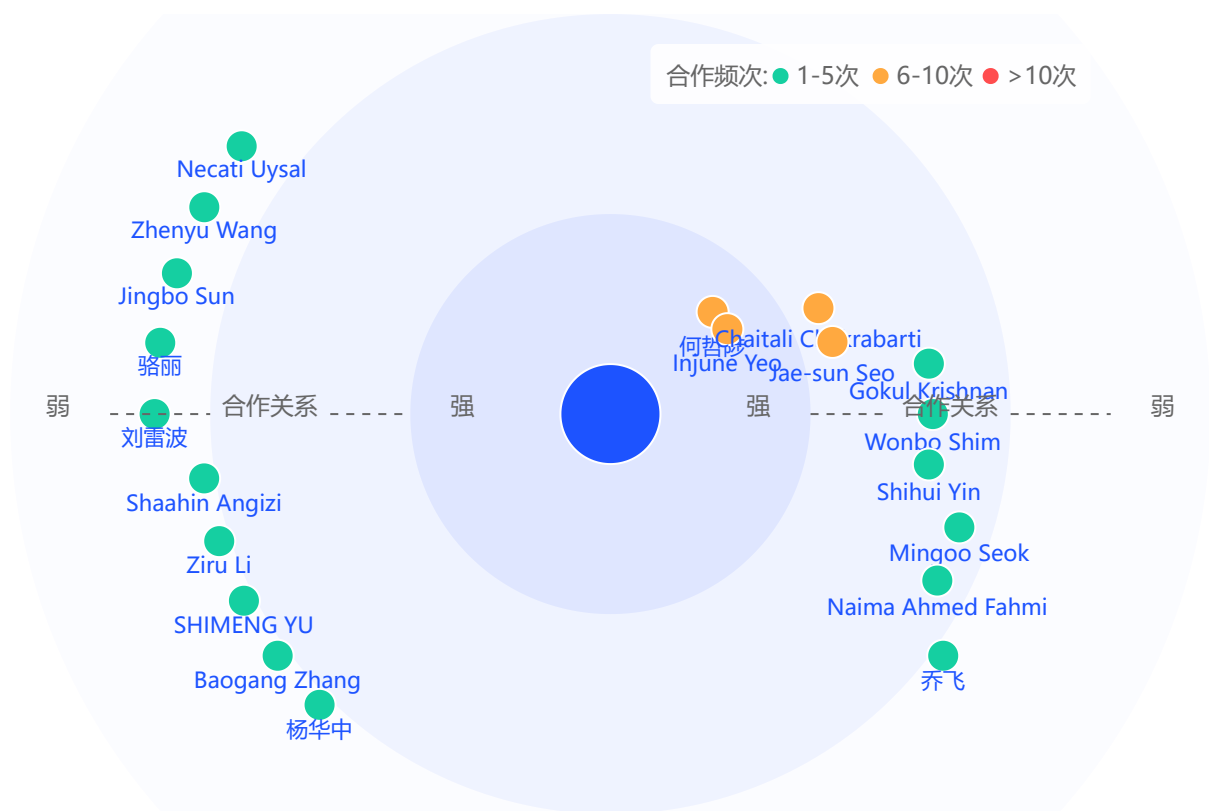
Exploring Model Stability of Deep Neural Networks for Reliable RRAM-Based In-Memory Acceleration

TC

Gokul Krishnan; Li Yang; Jingbo Sun; Jubin Hazra; Xiaocong Du; Maximilian Liehr; Zheng Li; Karsten Beckmann; Rajiv V. Joshi; Nathaniel C. Cady; Deliang Fan; Yu Cao

被引用次数: 0

学术合作



何哲陟	合作次数：8
上海交通大学	
Injune Yeo	合作次数：8
亚利桑那州立大学	
Chaitali Chakrabarti	合作次数：6
亚利桑那州立大学	
Jae-sun Seo	合作次数：6
康奈尔大学	
Gokul Krishnan	合作次数：4
亚利桑那州立大学	

还有 24 个合作者